AI-Powered Lung Disease Detection: An Explainable Dual-Modality System for Chest X-ray and CT Screening

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***Abstract*—Most deep-learning tools for chest imaging handle a single modality, return a label without an explanation, and will confidently label any image they are given. We describe a web-based system in which transfer-learned DenseNet-121 networks sort radiographs among COVID-19, Normal, Pneumonia and Tuberculosis, and sort CT slices among Benign, Malignant and Normal; the system explains every prediction, and refuses inputs it was not trained to handle. Before a classifier runs, each upload passes image validation, a per-modality sanity gate that rejects non-scans, a modality gate that rejects the wrong scan type, and a feature-space out-of-distribution detector; X-rays are also cropped to the lung fields by a U-Net. Each result carries a Grad-CAM heatmap, a non-diagnostic plain-language report from a large language model, and a scoped chat assistant, inside an application with verified accounts, scan history and rate limiting. A reproduction check can re-verify every checkpoint against its validation data before it is served. Data audits removed near-duplicate CT slices that had inflated test accuracy to 100% and burned-in artifacts in tuberculosis images. Held-out test accuracy is 96.57% for X-ray and 98.06% for CT. Sending the same X-ray test split through the deployed route exposed a preprocessing mismatch; after fixing it, the live route delivers 91.05%. External validation shows the limits of these figures: accepted external X-rays are classified at 93.9%, but the CT classifier misclassifies every external Normal slice, which further experiments trace to dataset-specific shortcut learning.**

***Index Terms*—chest X-ray, computed tomography, explainable AI, external validation, Grad-CAM, out-of-distribution detection, shortcut learning, transfer learning**

I. Introduction

Radiographs and computed tomography (CT) remain the main imaging routes by which clinicians identify tuberculosis, pneumonia, COVID-19 and lung cancer. Automated screening support matters especially in India, which carries about a quarter of the world's tuberculosis cases [1], where 80–85% of lung cancers are found at an incurable stage [2], and where there is roughly one radiologist per 100,000 people, concentrated in large cities [3]. Convolutional neural networks (CNNs) can match radiologists on narrowly defined chest-imaging tasks [4], [5], which has made automated triage an active area of applied research.

Three practical gaps limit how such models can be used. First, most systems handle one modality and give no explanation, so a clinician receives a label they cannot inspect. Second, a discriminative classifier assigns one of its known labels to *any* input, including photographs, screenshots and scans of the wrong type; in our own testing, a CT model labelled a screenshot "Malignant" with 93% confidence. Third, accuracy is usually reported offline on test images from the same sources as the training data, which says little about what a deployed system delivers or whether the model learned pathology rather than source-specific cues [6]–[8].

Problem statement. *To build one system that assigns a radiograph to one of COVID-19, Normal, Pneumonia or Tuberculosis and a CT slice to one of Benign, Malignant or Normal, explains every prediction, and refuses images it was not trained to handle, delivered as a secure multi-user web application whose reported accuracy is verified on the deployed code path and on external data.*

This paper presents such a system. Its contributions are:

1) A dual-modality classification pipeline using DenseNet-121 [9] with chest-radiograph pretraining for X-ray [10] and ImageNet pretraining for CT, with Grad-CAM heatmaps [11] and a safety-constrained LLM report and chat assistant.
2) An inference-time guard layer of image checks, a per-modality sanity gate, a modality gate and a feature-space out-of-distribution (OOD) detector, together with a reproduction check that re-verifies all six checkpoints against their validation data before they are served.
3) A data-quality audit trail (burned-in artifact cleanup, near-duplicate leakage detection with cluster-aware re-splitting, protocol-mismatch exclusion and a source-confound probe) that materially changed the reported results.
4) A live-serving audit that found and largely closed a 12–19-point train/serve accuracy gap, and an external validation that exposes, and experimentally diagnoses, shortcut learning in the CT classifier.
5) A complete web application with a FastAPI back end, a React front end that shows the guard checks to the user, verified accounts, scan history, rate limiting and a public deployment.

The remainder is organised as follows: prior work is surveyed in Section II; datasets and audits are covered in Section III; Section IV describes the proposed design; Section V covers how it was built and tested; Section VI presents results; and Section VII closes the paper.

TABLE I
POSITIONING AGAINST RELATED WORK

System	Two modalities	Input guards	Leakage audit	External test
CheXNet [4]	No	No	–	No
COVID-Net [12]	No	No	–	No
Rahman et al. [14]	No	No	–	No
Benamara et al. [17]	No	No	No	No
This work	Yes	Yes	Yes	Yes

TABLE II
FINAL DATA SPLITS AFTER CLEANING

Modality	Class	Train	Val	Test	Total
X-ray	COVID-19	2,856	357	357	3,570
	Normal	10,949	1,369	1,368	13,686
	Pneumonia	1,070	134	134	1,338
	Tuberculosis	498	62	63	623
CT	Benign	96	12	12	120
	Malignant	400	50	50	500
	Normal	333	42	41	416

II. RELATED WORK

Chest X-ray classification. CheXNet [4] showed that a DenseNet-121 trained on ChestX-ray14 detects pneumonia at a level comparable to radiologists, CheXpert [5] provided a large multi-label benchmark, and COVID-Net [12] applied a similar approach to COVID-19. On the datasets used here, Chowdhury et al. [13] screened viral and COVID-19 pneumonia and Rahman et al. [14] detected tuberculosis after lung segmentation, a design close to our U-Net crop. Mirugwe et al. [15] confirmed that transfer learning outperforms training from scratch for tuberculosis detection.

CT classification. CNNs are widely used to classify pulmonary nodules and lung cancer on CT [16]; on the same CT collection that we use, Benamara et al. [17] achieved an accuracy of 99.54%. Such studies typically use small single-institution datasets and rarely report a near-duplicate audit of their splits or an external test.

Spurious cues and unfamiliar inputs. According to Zech et al. [6], radiograph-based pneumonia classifiers learn to recognise which hospital produced an image, and their performance varies across sites; DeGrave et al. [7] found that COVID-19 detectors exploit source-specific shortcuts; Geirhos et al. [8] describe the phenomenon in general. Hendrycks and Gimpel [18] showed that softmax confidence is a weak signal for unfamiliar inputs, which our testing reproduced. Liu et al. [19] survey automatic report generation; our report layer instead uses a narrowly constrained prompt with a deterministic fallback.

Table I summarises the gap this work addresses.

III. DATA AND DATA-QUALITY AUDITS

Datasets. Radiographs are drawn from two public collections. The first [20] supplies the COVID-19, Normal and viral Pneumonia classes together with a lung mask for each image (its Lung Opacity class is not used) and the second [21] supplies Normal and Tuberculosis images. CT slices come from IQ-OTH/NCCD [22] (1,097 slices from 110 patients at one institution, exported as flat per-class JPEGs without patient IDs). The two X-ray sources were merged and 59 exact duplicates removed by SHA-256 hashing. Table II gives the final splits (seed 42). Both sets are imbalanced: Normal outnumbers Tuberculosis about 22 times, and Malignant outnumbers Benign about 4.2 times. Auxiliary data comprise 21,165 lung image–mask pairs for the U-Net and curated photographs, documents and screenshots as negatives for the sanity gates.

Burned-in artifacts. A scan of all 4,200 images of the TB source found a colour tint and/or a white redaction box in 53.1% of Tuberculosis images but only 0.37% of Normal images, a class-correlated shortcut. Both classes were cleaned uniformly: tinted images were desaturated, small boxes inpainted and images with large boxes excluded (72 of 700 TB images).

CT leakage. The first CT model, trained on a random per-image split, scored 100% on test. Perceptual hashing with structural-similarity (SSIM) confirmation [23] found 111 clusters of near-duplicate slices (287 pairs); 47 clusters spanned splits and 35 of 109 test images had a near-duplicate in training. Treating clusters as patient proxies, the data were resplit so that each cluster lies in one split (zero cross-split leaks on re-audit). A further audit found 61 Malignant slices from a different acquisition protocol (visible scanner table, different resolution) that were separable on protocol alone; they were excluded, leaving 1,036 slices.

Source confound. In the X-ray data, COVID-19 comes only from one database and Tuberculosis only from the other. Within the Normal class, which both databases contribute to, a logistic-regression probe on 32×32 thumbnails recovered the source database at 97.1% accuracy (AUC 0.996) on 3,495 images per source. Its effect on the classifier is analysed in Section VI-C.

IV. PROPOSED SYSTEM

A. Overview

The system is a three-tier web application: a React 19 single-page application, a FastAPI back end that owns six PyTorch models (two sanity gates, a modality gate, two disease classifiers and a U-Net), and a SQLite database, with the Groq LLM API, Gmail SMTP and Google OAuth as external services. The models are loaded once, on the first prediction request, and shared by all later requests. Fig. 1 shows the per-request prediction pipeline in execution order. Separate routes, `/predict/xray` and `/predict/ct`, make model selection a routing decision rather than a branch on a user-supplied field.

B. Disease Classifiers

Both classifiers use DenseNet-121 [9], whose feature reuse suits fine-tuning on datasets far smaller than Ima-

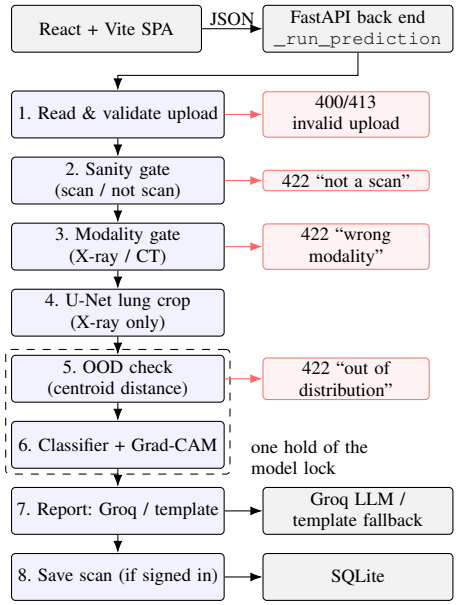
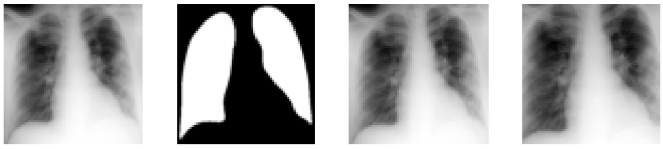


Fig. 1. Architecture and per-request prediction pipeline, top to bottom in execution order. Stages 1, 2, 3 and 5 can reject a request before the disease classifier runs. Both gates see the uncropped upload; only accepted X-rays are lung-cropped. Stages 5–6 run inside one hold of the classifier's lock.

geNet and whose roughly 8M parameters keep CPU inference practical. The X-ray model uses the TorchXRayVision `densenet121-res224-all` weights [10], pretrained on a large aggregate of chest radiographs, with a new four-class head and the library's single-channel preprocessing; it replaced an earlier ImageNet-pretrained model. The CT model uses ImageNet weights [24] with a new three-class head. Both are trained for 15 epochs (batch 32, seed 42): two epochs with the backbone frozen using Adam [25] at 10^{-3}, then full fine-tuning at 10^{-4}, with inverse-frequency class weights $w_c = N/(Kn_c)$ in the cross-entropy loss and flip, $\pm 10^\circ$ rotation and ± 0.1 brightness/contrast augmentation. The checkpoint with the best validation accuracy is kept (X-ray 96.46%; CT 99.04% at epoch 6).

C. Sanity Gate and Modality Gate

Both gates are ResNet-18 [26] binary classifiers. The two sanity gates (one per modality) separate real scans from photographs, documents and screenshots (X-ray: 15,373 positives vs. 7,460 negatives; CT: 829 vs. 3,000), reaching 99.96% and 100% validation accuracy. They run first because the modality gate is a closed-set classifier with no "neither" option: in testing, a random photograph was scored above 92% "X-ray" by the modality gate and also passed the OOD check. The modality gate, trained on the training splits relabelled by modality, reaches 100% validation accuracy; a request is rejected if it predicts the other modality or if its confidence is below 0.90. It was added because the OOD detector did not catch a real CT slice sent to the X-ray route.



(a) raw input (b) U-Net mask (c) seg. crop (d) border crop

Fig. 2. Lung-crop pipeline on a real test X-ray: (a) input; (b) U-Net mask; (c) segmentation crop; (d) border-crop fallback.

D. Out-of-Distribution Detector

The detector works in the classifier's own penultimate (post-pooling) feature space $\phi(\cdot)$. For each modality, the training set is embedded offline and the centroid and threshold are stored:

$$\mu = \frac{1}{N}\sum_{i=1}^{N}\phi(x_i), \qquad \tau = Q_{0.99}\big(\{\|\phi(x_i)-\mu\|_2\}_{i=1}^{N}\big). \quad (1)$$

An input is rejected if $\|\phi(x)-\mu\|_2 > \tau$, i.e., if it is farther from the training data than 99% of training images. The thresholds are 8.72 for X-ray (recomputed after the backbone change) and 25.62 for CT.

E. U-Net Lung Crop

The X-ray classifier was trained on lung-cropped images, and serving raw uploads changed its predictions (a genuine COVID-19 image was classified as Tuberculosis at 98.6% confidence). Accepted X-rays are therefore segmented by a 4-level U-Net [27] (32 base channels) at 256×256; the image is cropped to the mask's bounding box (4-pixel padding) and squash-resized to 224×224, as in training. If the mask is empty or its box covers less than 5% or more than 98% of the frame, a uniform 8% border crop is used instead (Fig. 2). The U-Net checkpoint was trained outside the project repository; its architecture was reconstructed from the checkpoint and confirmed by reproducing its validation Dice (Table III).

F. Explanation, Report and Chat

Grad-CAM [11] weights the last convolutional block's feature maps A^k by the spatially averaged gradients of the class score y^c, $\alpha_k^c = \frac{1}{Z}\sum_{i,j}\partial y^c/\partial A_{ij}^k$, and returns $\mathrm{ReLU}(\sum_k \alpha_k^c A^k)$, upsampled and alpha-blended (0.45) over the input. A new hook is registered per request. A low-confidence warning is attached when the top probability is below 0.50 (X-ray) or 0.55 (CT). The Groq-hosted `openai/gpt-oss-120b` model writes a two-to-three-sentence explanation from the class, confidence and probabilities under a prompt that forbids diagnosis, treatment advice and speculation; a fixed disclaimer is appended, and a template with the same disclaimer is used if the call fails. A chat assistant uses the same provider with a system prompt restricted to the tool and its output; it declines medical advice and off-topic requests, receives the current prediction and the last ten turns, and is capped at 2,000 characters per message.

Algorithm 1 Prediction pipeline (`_run_prediction`)

Input: upload f, route modality m, optional user u
Output: class, probabilities, heatmap, report; or HTTP error

if u signed in **and** unverified **then return** 403
end if
$I \leftarrow$ decode f ▷ 413 if >10 MB; 400 if invalid
if grey std < 10 **or** colour spread > 20 **then return** 400
end if
$x \leftarrow 224{\times}224$ ImageNet-normalised copy of I
if $\text{Sanity}_m(x) \neq$ scan **then return** 422
end if
$(\hat{m}, p) \leftarrow \text{ModalityGate}(x)$
if $p < 0.90$ **or** $\hat{m} \neq m$ **then return** 422
end if
if m = X-ray **then** $x \leftarrow \text{Prep}_{\text{XRV}}(\text{LungCrop}(I))$
end if
acquire lock of Classifier_m
if $\|\phi_m(x) - \mu_m\|_2 > \tau_m$ **then** release; **return** 422
end if
$(probs, heatmap) \leftarrow \text{Classifier}_m(x)$ with Grad-CAM
release lock; add warning if $\max(probs) < c_m$
$report \leftarrow$ LLM report, or template on failure
if u signed in **then** save scan for u
end if
return result

V. Implementation and Testing

A. Prediction Pipeline

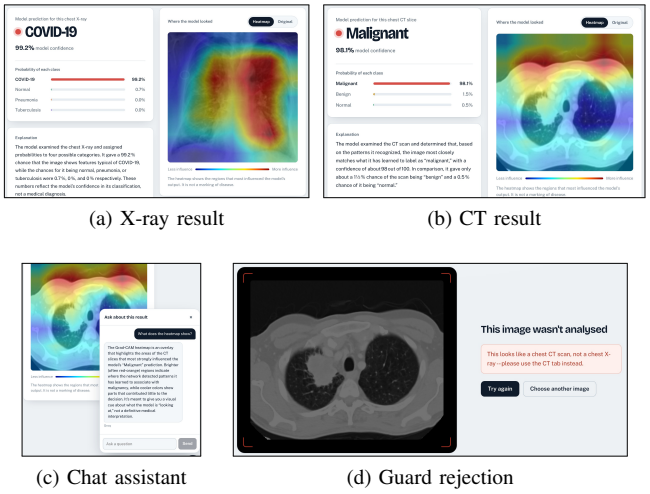
Algorithm 1 gives `_run_prediction()`, shared by both prediction routes. The OOD check and the classification with Grad-CAM run inside one continuous hold of a per-model lock: FastAPI runs synchronous handlers in a thread pool, and Grad-CAM's forward hooks on a shared model let an overlapping request observe another request's hooks, a failure seen during development.

B. Model Loading and Reproduction Check

The server starts without loading any model. The first prediction request loads all six checkpoints, the two OOD statistics files and the U-Net under a process-wide lock with a double check, so concurrent first requests wait for a single load and later requests reuse the loaded models. Each checkpoint stores its claimed validation metric. While a checkpoint is loaded, a reproduction check recomputes that metric on the validation split with the same preprocessing and refuses to serve the model if the difference exceeds a tolerance. With the strict tolerances of 0.1 percentage point (accuracy) and 1 point Dice / 2 points IoU (U-Net), Table III shows a verified load. The check caught a real checkpoint/data mismatch during development. It needs the validation data on disk and took two to three minutes per model on a CPU, so the tolerances are configurable: the current public configuration sets them to 1.0, which skips the check and makes the first prediction fast, and the strict values are restored for verification runs.

TABLE III
Reproduction Check (Strict Tolerances)

Model	Claimed	Reproduced
X-ray classifier	acc 0.9646	acc 0.9646
CT classifier	acc 0.9904	acc 0.9904
Modality gate	acc 1.0000	acc 1.0000
X-ray sanity gate	acc 0.9996	acc 0.9996
CT sanity gate	acc 1.0000	acc 1.0000
U-Net	Dice 0.989, IoU 0.979	Dice 0.983, IoU 0.967



(a) X-ray result (b) CT result

(c) Chat assistant (d) Guard rejection

Fig. 3. Screens of the deployed application (light theme): (a) COVID-19 prediction for a chest X-ray and (b) Malignant prediction for a CT slice, each with class-probability bars, Grad-CAM heatmap and explanation; (c) the chat panel explaining a heatmap; (d) a CT slice uploaded on the X-ray tab, refused by the modality gate with the reason.

C. Application, Security and Deployment

The back end exposes fifteen routes: two prediction routes, nine authentication routes, chat, scan history and two status routes. Passwords are hashed with bcrypt; sessions are HS256 JSON Web Tokens valid for seven days; email-verification (24 h) and password-reset (1 h) links are single-use tokens sent over Gmail SMTP. Google sign-in uses a signed `state` bound to a nonce cookie, and email links and redirects follow the user's origin only if it is allow-listed. A signed-in user whose email is unverified receives HTTP 403 from prediction and chat. Per-client limits (20/min for prediction, chat and login; 30/h for sign-up; 5/h for password reset) use the real client address, trusting `X-Forwarded-For` only from the local proxy. SQLite stores users, tokens and scans; each scan keeps its probabilities, heatmap and report, so the history page reproduces what the user saw. The React front end separates public and protected routes and shows unverified users only a verification screen. It defaults to a dark theme, with a light theme available. The landing page animates the four guard checks on a photograph and a chest X-ray and shows internal against external accuracy. An analysing screen steps through a four-stage checklist. The result page shows the class, its confidence, per-class probability bars, the Grad-CAM heatmap

TABLE IV
HELD-OUT TEST PERFORMANCE (OFFLINE CLASSIFIER)

	Class	Prec.	Recall	F1	n
X-ray (acc. 96.57%)	COVID-19	0.926	0.944	0.935	357
	Normal	0.982	0.972	0.977	1,368
	Pneumonia	0.913	0.940	0.927	134
	Tuberculosis	0.955	1.000	0.977	63
	Macro avg.	0.944	0.964	0.954	1,922
CT (acc. 98.06%)	Benign	1.000	0.833	0.909	12
	Malignant	1.000	1.000	1.000	50
	Normal	0.953	1.000	0.976	41
	Macro avg.	0.984	0.944	0.962	103

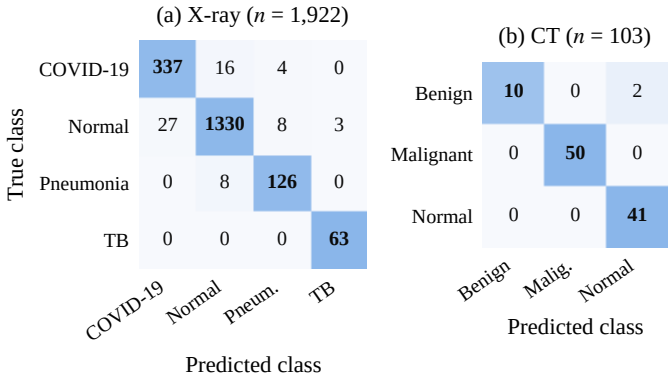


Fig. 4. Confusion matrices on the held-out test splits (counts; shading by row-normalised rate).

with a toggle to the original image and the explanation, labelled as written by a language model, with a chat panel for questions about the result. A rejection appears as "This image wasn't analysed" with the reason and options to try again or choose another image, and a History page lists saved scans as heatmap cards (Fig. 3). For public access, a launch script (PowerShell on Windows, Bash on macOS) builds the front end, starts the back end on the loopback interface, serves the build with `vite preview` (which proxies `/api` to the back end) and exposes it through an ngrok HTTPS tunnel.

D. Testing

A 61-test `pytest` suite starts the real application with a temporary database and mocked SMTP and LLM clients; the first prediction test loads all six models. It covers predictions on real images, history saving, the HTTP 403 returned to unverified users by the prediction and chat routes, wrong-modality, blank, colour, non-image and empty uploads, the uncertain-modality guard, authentication, origin handling, OAuth state tampering, client-IP extraction and chat limits; all tests pass. Over live HTTP, wrong-modality scans, photographs, screenshots, documents, blank, noise and corrupt files were all rejected with the intended status codes. Twelve simultaneous predictions matched their sequential results exactly. Mean end-to-end request latency was 1.26 s (X-ray) and 1.29 s (CT) over eight sequential requests each on the development laptop.

VI. RESULTS AND DISCUSSION

A. Held-out Test Performance

Table IV and Fig. 4 report the classifiers on their held-out test splits. Validation and test agree (X-ray 96.46% vs. 96.57%; CT 99.04% vs. 98.06%). The CT figure was obtained after the leakage fix of Section III; its only errors are two Benign slices predicted Normal, and with 12 Benign test images each error moves Benign recall by 8.3 points. These are in-distribution results.

B. Live-Serving Accuracy and the Crop Fix

Sending the test images through the deployed `/predict/xray` route revealed live accuracy 12–19 points below the offline figure, with predictions drifting to COVID-19: on a class-balanced sample, 80% of served images (350/436) were predicted COVID-19. The cause was the crop function, which could return a non-square box that the X-ray preprocessing then centre-cropped, removing lung margins that every training image had kept. Making every crop path end in the training squash-resize closed most of the gap (Table V). With the fix, the full 1,922-image test split through the live API with all guards active gives 91.05% (rejections counted as errors; 91.67% over the 1,909 served images, 13 rejected by the OOD detector). The CT route shows no loss (98.06% live). The remaining X-ray gap, mainly Normal predicted as COVID-19 (82/1,368), comes from a class-dependent crop difference in the training data (mask crops for COVID-source classes, border crops for TB-source classes) and needs a rebuilt training set to close. The reproduction check cannot detect such defects, because it replays the checkpoint on pre-processed validation data; only testing the real route did. The U-Net crop also trades accuracy across classes: on a 60-image sample it raised overall accuracy from 88.3% (border crop) to 95.0% but lowered Tuberculosis from 14/15 to 13/15.

TABLE V
X-RAY ACCURACY BEFORE AND AFTER THE CROP FIX

Measurement	Offline	Live before	Live after
Original uploads (weighted)	0.9657	0.8833	0.9500
Pre-cropped split (simulated)	0.9667	0.8000	0.9208
Served predicted COVID-19	–	80%	27%
Full test split, live API	–	–	0.9105

C. Source Confound in the X-ray Classifier

Two follow-ups to the source probe, run on the earlier ImageNet-backbone X-ray model, tested whether the classifier exploits the source signal. Blanking the central lung region left COVID-19 accuracy almost unchanged (0.919 to 0.902) while Pneumonia (0.978 to 0.657) and Normal (0.993 to 0.824) collapsed, and COVID-19 was the only class whose Grad-CAM attention fell outside the lungs more often than a geometric baseline (75.6% vs. 68.4%). COVID-19 predictions therefore appear partly keyed to source characteristics, consistent with

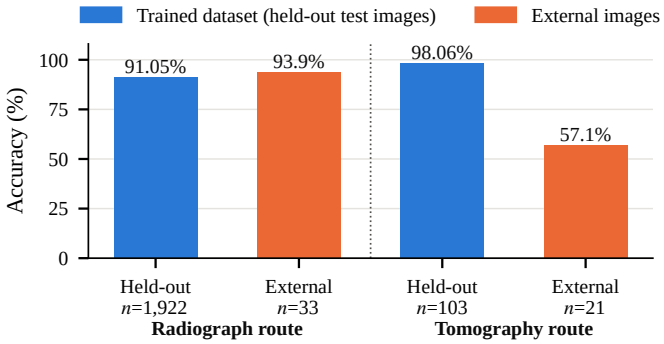


Fig. 5. Accuracy through the deployed routes on the trained datasets (held-out test images from the datasets the models were trained on) and on external images. X-ray external accuracy is over the images accepted by the guards.

[7]. The analysis has not yet been repeated on the current backbone.

D. External Validation

Fig. 5 compares internal and external accuracy through the deployed routes with all guards active. On 40 external X-rays (10 per class), accepted images were classified at 93.9% (31/33), but the guards rejected 7 genuine radiographs (3 OOD; 4 tinted TB images by the colour check), giving 77.5% end to end. The sample is small and its provenance relative to the training databases could not be verified, so it shows only that the X-ray model does not collapse.

On CT the result is clear: accuracy fell from 98.06% to 57.1% (12/21), and none of the nine external Normal slices was classified Normal (seven Malignant, two Benign); no external Benign source was available. Grad-CAM on the failures showed attention on the mediastinum and great vessels rather than the lung parenchyma, suggesting the Normal class was learned from one dataset's slice-selection and windowing style. Offline, the deployed CT classifier labelled 0 of 264 external Normal slices (53 patients, two independent sources) as Normal. A retraining experiment that added anatomically diverse external Normal slices recovered 250/264 on held-out external Normal patients, but flipped all 12 external Malignant slices from correct to Normal; an ablation that excluded the Malignant set's source from training gave the same 0/12. Both classes therefore rode on the same acquisition-style shortcut, and the original 12/12 on external Malignant was the shortcut pointing the other way rather than evidence of lesion detection. The retrained model was not adopted. A volumetric pipeline (3D nodule detector and nodule classifier on DICOM series) was also prototyped and withdrawn after reaching 28.6% (4/14) on the independent LUNGx cohort. We therefore treat the CT route as a proof of concept whose accuracy is specific to its source dataset; neither the test split nor the reproduction check could have revealed this.

E. Limitations

The CT dataset is small and single-institution, and the CT classifier does not generalise beyond it. External X-ray results rest on 40 images, and the source confound has not been re-measured on the current backbone. The gates were trained on cropped images but receive uncropped uploads, and the OOD detector does not reject genuine chest images of conditions outside the label set (in testing, Lung Opacity X-rays were assigned to the trained classes). The U-Net has no training script in the repository. In the current fast-start configuration the reproduction check is skipped, so a checkpoint that no longer matches its data would be served until a strict-mode run is repeated. The prediction and chat routes accept anonymous calls bounded only by rate limits, sessions cannot be revoked before expiry, rate limiting is per process and SQLite is single-writer, so a multi-server deployment would need changes; throughput under load was not measured.

VII. Conclusion and Future Work

We presented a dual-modality system that classifies chest X-rays and CT slices with transfer-learned DenseNet-121 networks, explains every prediction with Grad-CAM and a non-diagnostic language-model report, refuses non-scans, wrong-modality and unfamiliar inputs through a guard layer, and can verify that every served checkpoint reproduces its validation metrics, inside a complete multi-user web application. Held-out accuracy is 96.57% (X-ray) and 98.06% (CT), and the deployed X-ray route delivers 91.05%. The more transferable result is methodological: auditing the data corrected an inflated CT accuracy, testing the real route found a train/serve mismatch, and external validation with a controlled retraining experiment showed that the CT classifier relies on an acquisition-style shortcut that no in-distribution check could reveal. Future work includes multi-institution CT data with patient identifiers and external-source data for every class, a larger provenance-checked external evaluation, rebuilding the X-ray training data with one crop convention, training the gates on uncropped inputs, an open-set mechanism for untrained findings, confidence calibration, a clinical reader study, and production hardening with mandatory authentication, revocable sessions, a reproduction check that runs in the background so it can stay enabled without delaying the first prediction, and continuous integration.

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